

**Experiential Learning
Project Assessment-I on
“ Password Strength
Analyzer Using Hashing”
under the course**

Python ProgrammingLanguage(N-PCCCM304P)

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**Under the Guidance
of
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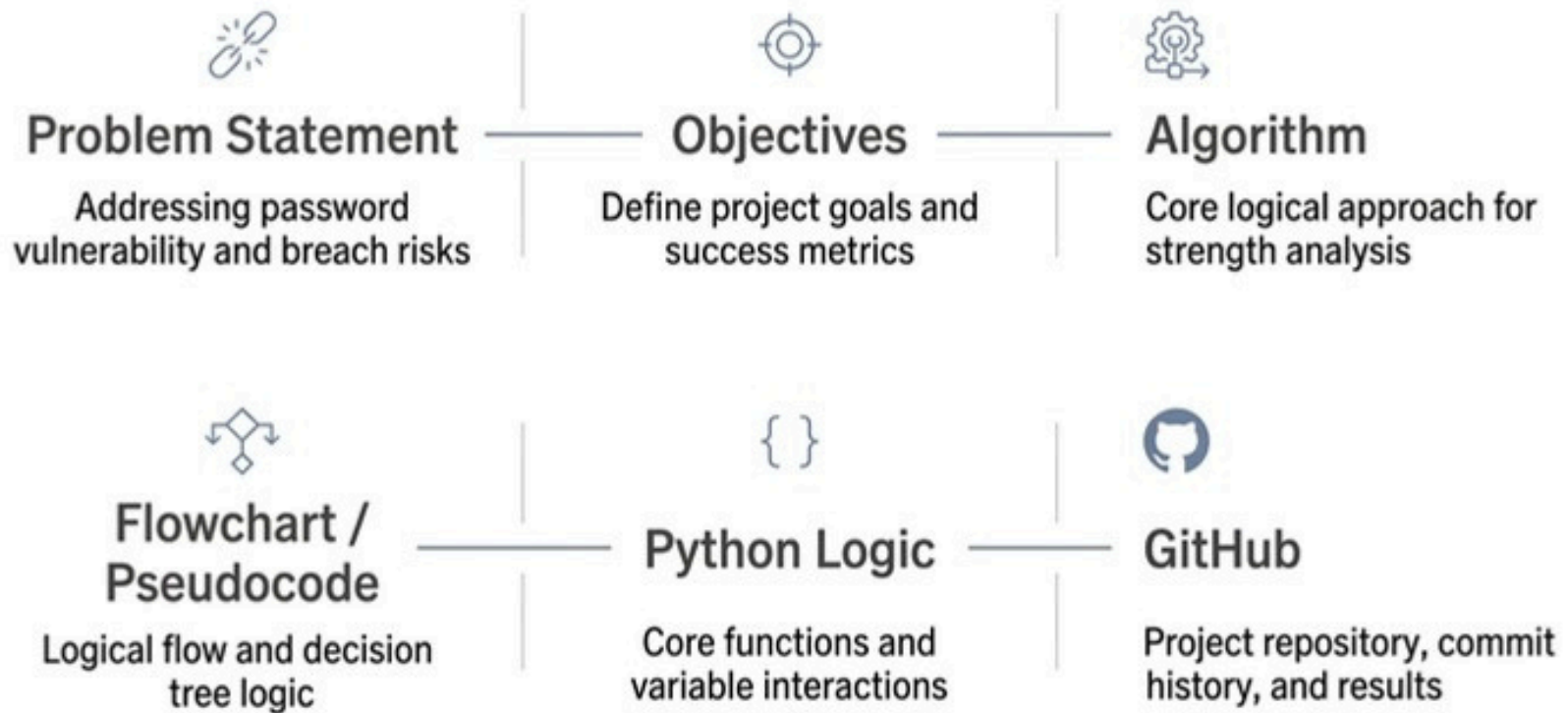
Session:2026-27 (ODD)

Department of CSE(AI&ML)

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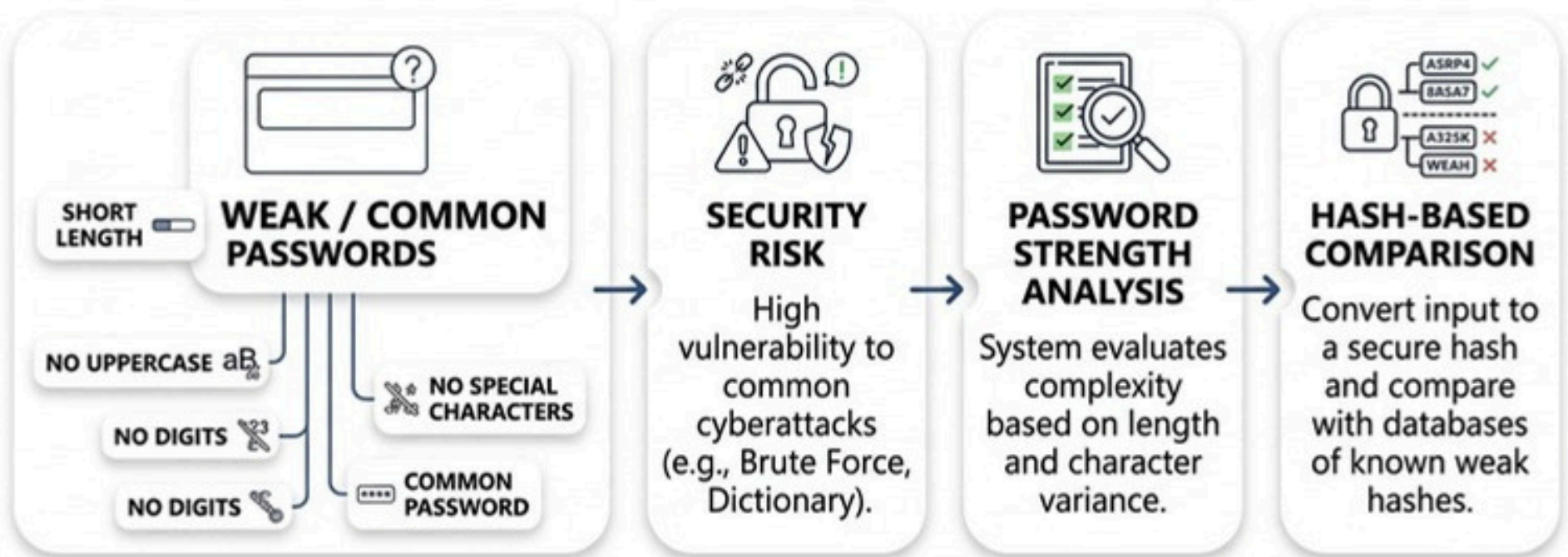
PROJECT OVERVIEW



PROBLEM STATEMENT & OBJECTIVES:

- **Password Strength Analyzer Using Hashing**

- Many users choose weak or commonly used passwords, creating security risks.
- The project analyzes password strength and uses hashing for security evaluation.



"Evaluate password security using hashing techniques."

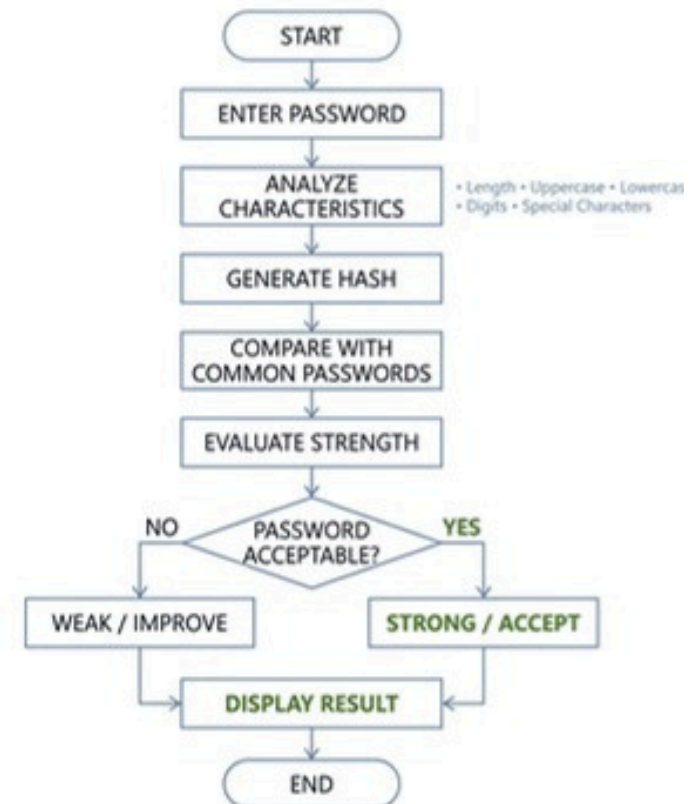
ALGORITHM & FLOWCHART

- The algorithm defines the step-by-step process from password input to the final result.
- It includes checking characteristics, hashing, comparison, and strength evaluation..

ALGORITHM

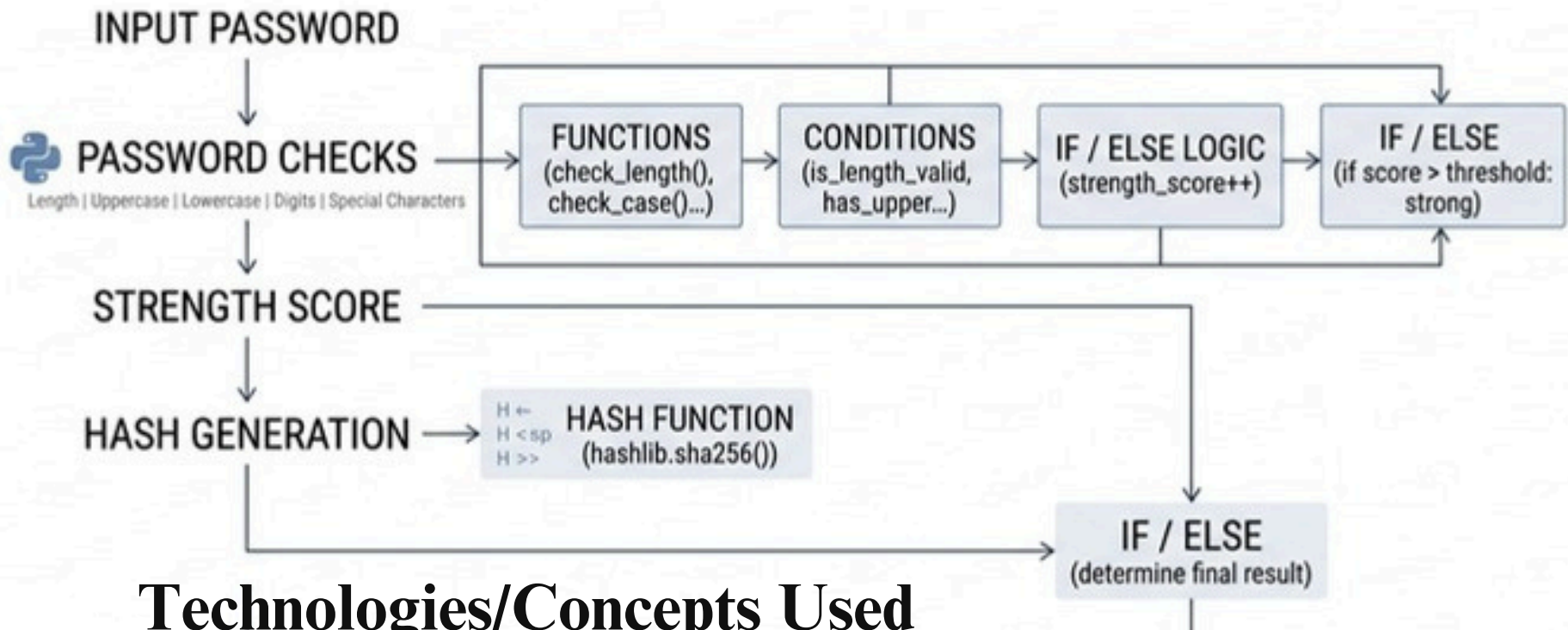
- | | |
|----|--------------------------------|
| 01 | Input Password |
| 02 | Check Password Characteristics |
| 03 | Generate Hash |
| 04 | Compare with Common Passwords |
| 05 | Evaluate Strength |
| 06 | Display Result |

FLOWCHART



UNDERSTANDING OF PYTHON LOGIC

- Python is used for input, conditions, password checking, scoring, and hashing.
- The program uses if/else, functions/logic, and hashlib to generate the result. passwords



Technologies/Concepts Used

- Python
- Conditions
- Functions/logic
- hashlib
- SHA-256

PHASE-II: CODING & IMPLEMENTATION

Technology Used

Python

Implementation

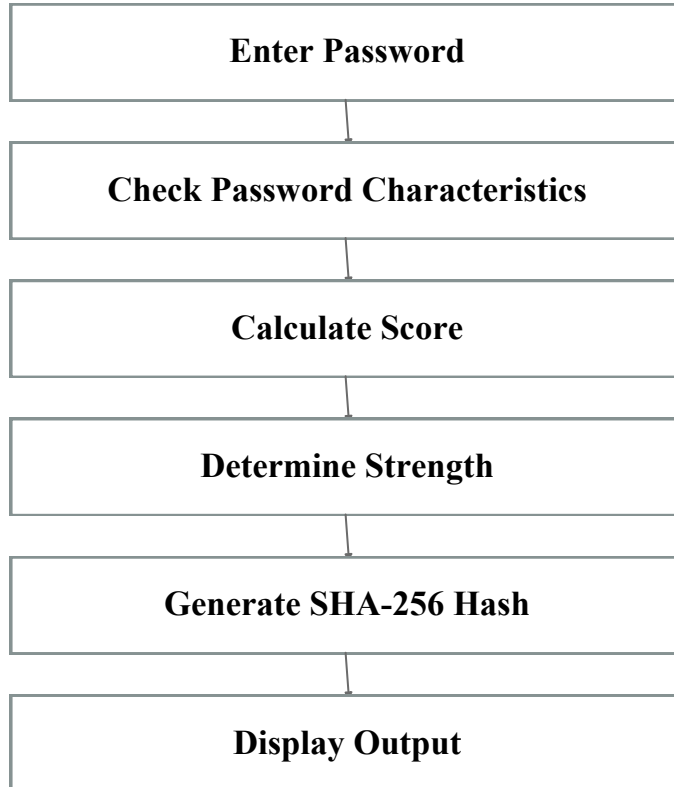
- Password is taken as user input.
- Password length is checked.
- Uppercase and lowercase characters are checked.
- Digits and special characters are checked.
- A score is calculated based on the conditions satisfied.
- Password strength is classified as Weak, Medium, or Strong.
- SHA-256 hashing is performed using the hashlib library.



```
Python Code
import hashlib.py U
> Users > RUCHIKA > Downloads > oraclejdk-26 > import hashlib.py > check_password
1 import hashlib
2
3 def check_password(password):
4     score = 0
5
6     if len(password) >= 8:
7         score += 1
8     if any(c.isupper() for c in password):
9         score += 1
10    if any(c.islower() for c in password):
11        score += 1
12    if any(c.isdigit() for c in password):
13        score += 1
14    if any(not c.isalnum() for c in password):
15        score += 1
16    return score
17    [ Insert Python code screenshot here ]
18 def get_strength(score):
19     if score <= 2:
20         return "Weak"
21     elif score <= 4:
22         return "Medium"
23     else:
24         return "Strong"
25
26 def generate_hash(password):
27     return hashlib.sha256(password.encode()).hexdigest()
28
29 password = input("Enter password: ")
30
31 score = check_password(password)
32 strength = get_strength(score)
33 password_hash = generate_hash(password)
34
35 print("Password Strength:", strength)
36 print("Hash:", password_hash)
0 Indexing completed.
```

PHASE-II: PROGRAM OUTPUT

Working Flow



Sample Output

```
Enter password: apple
Password Strength: Weak
Hash: 3a7bd3e2360a3d29eea436fcfb7e44c735d117c42d1c1835420b6b9942dd4f1b
PS C:\Users\RUCHIKA\AppData\Local\Programs\Microsoft VS Code> & C:\Users\R
Enter password: C:\Users\RUCHIKA\AppData\Local\Programs\Python\Python313\p
Password Strength: Strong
Hash: 00c2c199863502f881c5a3e0d02b5e29214f85350fc11eb41742166cbaa80a40
PS C:\Users\RUCHIKA\AppData\Local\Programs\Microsoft VS Code> & C:\Users\R
Enter password: tueijf
Password Strength: Weak
Hash: daa848bb46f09af7d2c920b539c2bdadaf9aa499292096f5de34ec15664d8a4c
```

Implemented Features

- Strength classification
- Score-based evaluation
- SHA-256 hash generation
- Console output



Thankyou

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